Novartis Foundation Symposium
Edited by David Evered, Julie Whelan
Copyright © 1988 by Ciba Foundation

Distinct roles of neurofilament and tubulin gene expression in axonal growth

Paul N. Hoffman

Departments of Ophthalmology and Neurology, The Johns Hopkins School of Medicine, Baltimore, Maryland 21205, USA

Abstract. Tubulin and the neurofilament (NF) proteins, which are major constituents of the axonal cytoskeleton, play distinct roles in longitudinal and radial growth of axons. The predominantly longitudinal growth of axons in developing neurons correlates with relatively high levels of expression for a particular tubulin gene, encoding the class II beta tubulin isotype, whereas levels of gene expression for two other beta tubulin isotypes (classes I and IV) are comparable in developing and maturing neurons. Gene expression for the 68 kDa NF protein (NF68) is low during longitudinal growth. Conversely, relatively high levels of gene expression for NF68 and low levels for class II beta tubulin in maturing neurons correlate with the radial growth of axons. The developmental pattern of gene expression is recapitulated during axonal regeneration. Expression of the class II beta tubulin isotype correlates with the outgrowth (elongation) of regenerating sprouts.

1988 Plasticity of the neuromuscular system. Wiley, Chichester (Ciba Foundation Symposium 138) p 192–204

The potential for axonal growth is expressed in mature neurons during axonal regeneration. When an axon in the mammalian peripheral nervous system is interrupted by cutting or crushing a nerve (axotomy), the distal stump degenerates, and is replaced by the outgrowth (elongation) of regenerating sprouts arising from the proximal stump (Ramón y Cajal 1928). It is generally assumed that developmental mechanisms for axonal growth are recapitulated during axonal regeneration. The principal cytoskeletal proteins of the axon, tubulin and the neurofilament (NF) proteins, are directly involved in the process of axonal growth (Lasek & Hoffman 1976). In this review I shall consider the distinct roles of these cytoskeletal proteins in longitudinal and radial growth of axons. In addition I shall examine recent evidence indicating that the developmental pattern of gene expression for these proteins is recapitulated during axonal regeneration.

Axonal transport of the cytoskeleton

The axonal cytoskeleton is a three-dimensional network consisting of microtubules (MT), NF, and cross-linking structures (Hirokawa 1982). NF are heteropolymers composed of three distinct protein subunits (with molecular weights of 68, 145 and 200 kDa, respectively) (Hoffman & Lasek 1975). Tubulin, the major subunit protein of MT, is a dimer comprising one alpha and one beta tubulin subunit. Since the axon is devoid of ribosomes, axonal proteins are synthesized in the neuron cell body (soma) before undergoing somatofugal translocation within the axon via the axonal transport system. Tubulin and the NF proteins are transported at velocities of 1–2 mm/day in the slow component of axonal transport (Hoffman & Lasek 1975). NF proteins appear to be assembled in the cell body shortly after synthesis and undergo axonal transport in the form of NF polymers (Black & Smith 1988); axons do not contain a significant pool of unassembled NF protein (Morris & Lasek 1982). In contrast, it is likely that both assembled MT and unassembled tubulin subunits undergo axonal transport (Brady et al 1984, D.F. Watson, J.W. Griffin & P.N. Hoffman, unpublished observations). The transport of unassembled tubulin provides the potential for the assembly of MT in the distal axon.

Cytoskeletal gene expression

The mRNAs encoding tubulin and the NF proteins are transcribed from genes located in the cell nucleus and translated into proteins on ribosomes located in the neuron cell body. Each of the NF proteins is encoded by a separate gene (Lewis & Cowan 1985, Julien et al 1986). The situation is more complex for tubulin; alpha and beta tubulin are each encoded by multigene families (Sullivan & Cleveland 1986). Within these families distinct tubulin isotypes can be distinguished on the basis of their unique C-terminal amino acid sequences. There are five beta tubulin isotypes; each is encoded by a separate gene (Sullivan & Cleveland 1986). Genes encoding three of these isotypes (classes I, II and IV) have been cloned from rat (Bond et al 1984, Farmer et al 1984, Sullivan & Cleveland 1986). *In situ* hybridization studies with specific cDNA probes demonstrate that each of these beta tubulin isotypes is expressed in rat sensory neurons (Hoffman & Cleveland 1988).

Recently, we examined the differential expression of genes encoding these beta tubulin isotypes (classes I, II and IV) and the 68 kDa NF protein (NF68) in rat sensory neurons during development, maturation, and axonal regeneration (Hoffman & Cleveland 1988). In developing neurons of five-day-old animals axonal growth is primarily longitudinal; significant radial growth has not yet begun (N.A. Muma, E.H. Koo & P.N. Hoffman, unpublished

10.1002/9780470513675.ch12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/9780470513675.ch12 by Sichuan University, Wiley Online Library on [02/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

observations). Conversely, in maturing neurons of 10-week-old animals longitudinal growth is largely completed (Martinez & Friede 1970) and radial growth predominates (Hoffman et al 1985a). Expression of the specific tubulin gene encoding the class II isotype was induced to high levels during both development and axonal regeneration, whereas the expression of genes encoding the two other isotypes (classes I and IV) remained comparable to mature levels. Conversely, expression of NF68 was relatively low during both development and regeneration. In maturing neurons, levels of gene expression were relatively high for NF68 and low for class II beta tubulin. Thus, inverse patterns of gene expression for NF68 and the class II tubulin isotype were found in developing and maturing neurons. The development pattern was recapitulated in maturing neurons during axonal regeneration.

Tubulin expression and axonal growth

The expression of a specific tubulin gene, encoding a distinct beta tubulin isotype (class II), is selectively induced during neuronal development and axonal regeneration. It is not surprising that tubulin is expressed at high levels during development and regeneration because MT are the principal cytoskeletal elements in both developing axons and regenerating sprouts (Berthold 1978). It has been suggested that the ability of MT to undergo reversible assembly and disassembly contributes to the plasticity of developing axons (Lasek 1981).

Tubulin participates directly in the mechanism of axonal elongation (Yamada et al 1971) and the delivery of tubulin by slow axonal transport may limit the rate at which axons elongate during regeneration (Lasek & Hoffman 1976, Wujek & Lasek 1983). Increased expression of the class II isotype after axotomy correlates with increases in regenerating nerve fibres in both the total amount of pulse-labelled tubulin undergoing axonal transport and the relative amount of this tubulin conveyed in the faster phase of slow axonal transport (SCb) (Hoffman & Lasek 1980, Hoffman et al 1985b). Thus, increased amounts of class II beta tubulin may be transported in SCb in regenerating nerve fibres, thereby increasing the delivery of this isotype to regenerating sprouts. This raises the possibility that the class II beta tubulin isotype plays a special role in the outgrowth of regenerating sprouts.

Morphological correlates of NF expression

NF are major intrinsic determinants of axonal calibre in myelinated nerve fibres, where NF number correlates directly with the axonal cross-sectional area (Friede & Samorajski 1970, Weiss & Mayr 1971, Berthold 1978, Hoffman et al 1984). The low level of NF expression is consistent with the observed paucity of axonal NF in developing neurons (Peters & Vaughn

10.1002/9780470513675.ch12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/9780470513675.ch12 by Sichuan University, Wiley Online Library on [02/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

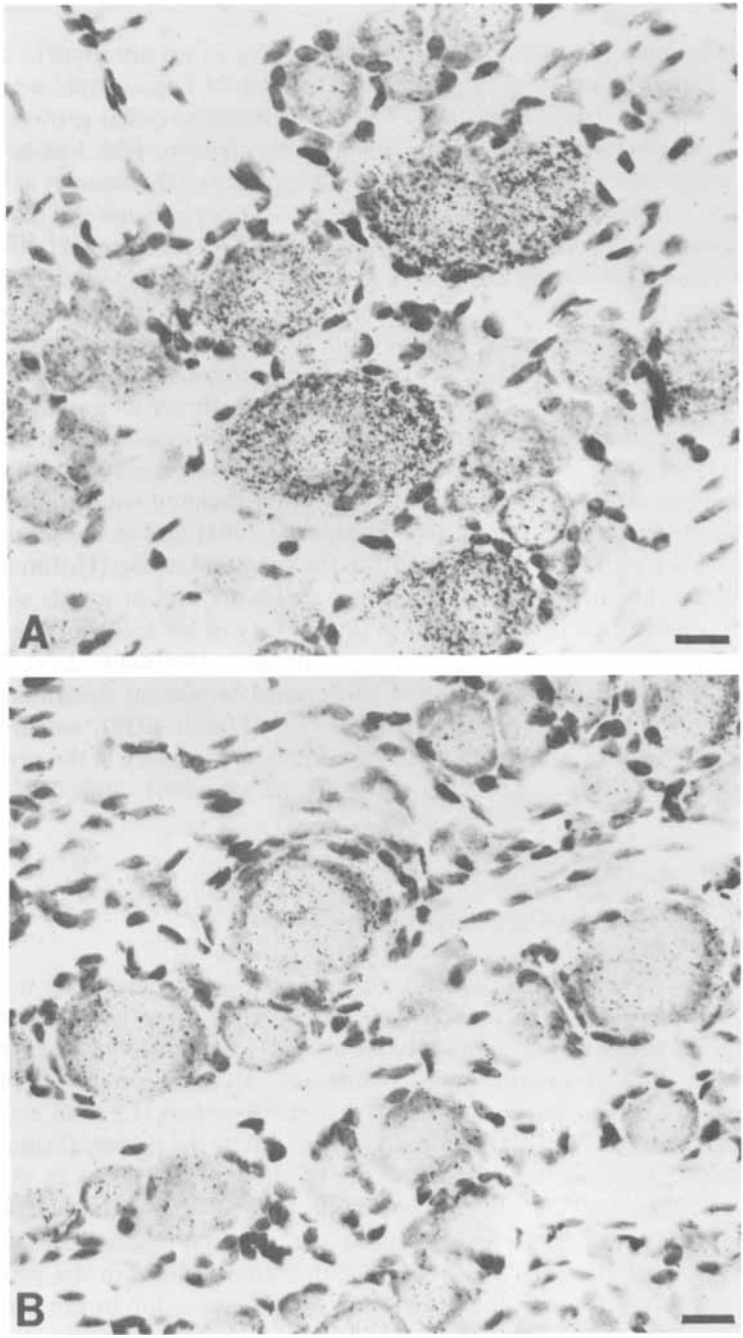


FIG. 1. Reduced NF68 gene expression after axotomy. ^{35}S-labelled NF68 cDNA was hybridized *in situ* with control (A) and axotomized lumbar sensory neurons of rat, two weeks after crushing the sciatic nerve (B). Levels of NF68 mRNA, as reflected by the density of silver grains in these autoradiograms, are normally greater in large than in small sensory neurons (A). After axotomy, mRNA levels are reduced in both large and small neurons (B). Bars, 10 μm. (Reproduced from Hoffman et al 1987).

10.1002/9780470513675.ch12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/9780470513675.ch12 by Sichuan University, Wiley Online Library on [02/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

1967). Before undergoing radial growth, developing axons are small in calibre (less than 1mm in diameter), are highly enriched in MT, and contain few NF (Peters & Vaughn 1967, Berthold 1978). In mammals, radial growth is restricted to myelinated axons. Axons arising from neurons with low levels of NF gene expression remain small throughout ontogeny (Hoffman et al 1987), and include normal unmyelinated fibres which are morphologically indistinguishable from embryonic axons (Berthold 1978). The induction of NF gene expression during postnatal development correlates with the radial growth of myelinated nerve fibres (N.A. Muma, E.H. Koo & P.N. Hoffman, unpublished observations), a process which begins at approximately 10 days of age (N.A. Muma, E.H. Koo & P.N. Hoffman, unpublished observations) and continues until at least six months of age in rat (Hoffman et al 1985a).

The decrease in axonal calibre which occurs in the proximal stump after axotomy (Greenman 1913) correlates with reduced NF gene expression (Fig. 1) (Hoffman et al 1987). This axonal atrophy is associated with proportional reductions in axonal NF content (Hoffman et al 1984) and in the amount of NF protein undergoing axonal transport in the proximal stump (Hoffman et al 1985b). Since this atrophy begins near the cell body and proceeds somatofugally along the fibre at a rate equal to the velocity of NF transport, we have termed this process somatofugal axonal atrophy Hoffman et al 1987). Moreover, since axonal calibre is the single most important determinant of conduction velocity in myelinated nerve fibres (Hursh 1939), somatofugal axonal atrophy is associated with reduced conduction velocity in the proximal stumps of regenerating nerve fibres (Cragg & Thomas 1961, Gillespi & Stein 1983).

Regulation of cytoskeletal gene expression

Maintenance of the pattern of gene expression found in maturing neurons (i.e., high for NF and low for class II beta tubulin) may be regulated by neuron–target interactions. Somatofugal atrophy, the morphological correlate of reduced NF gene expression (Hoffman et al 1987) is initiated when a sensory neuron is disconnected from its peripheral targets (Czéh et al 1977). If the sciatic nerve is cut and regeneration prevented, the proximal stumps of axons remain atrophic throughout the life of the animal (Weiss et al 1945, Dyck et al 1984). On the other hand, if the nerve is crushed and effective regeneration allowed, calibre recovers in the proximal stump (Kuno et al 1974, Hoffman et al 1984). The recovery of normal calibre in the proximal axon, which correlates with the restoration of NF expression to pre-axotomy levels, coincides with the reinnervation of target tissues (Kuno et al 1974). Thus, specific interactions between neurons and their targets may play important roles in both the induction of NF gene expression and the down-

10.1002/9780470513675.ch12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/9780470513675.ch12 by Sichuan University, Wiley Online Library on [02/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

regulation of class II beta tubulin expression. The transition from the developmental to the maturational pattern of gene expression at approximately 10 days of age in rat sensory neurons (N.A. Muma, E.H. Koo & P.N. Hoffman, unpublished observations) coincides in a general manner with the time when sensory axons establish their mature patterns of peripheral innervation (Diamond 1981).

An unresolved issue is the extent to which neuron–target interactions specify the level of NF gene expression. Each class of target cells is innervated by axons of a specific calibre. For example, extrafusal muscle fibres and encapsulated sensory receptors are innervated by large myelinated axons, intrafusal muscle fibres are innervated by small myelinated axons, and free nerve endings, which lack specific receptors, arise from small unmyelinated sensory fibres (Brodal 1981). This relationship between the physiological function of targets and axonal calibre raises the possibility that target cells help to specify the level of NF gene expression, and thereby determine axonal calibre. According to this hypothesis, the quantity or quality of target-induced signal would determine the rate of NF gene expression, and each class of target cells would specify characteristic levels of signal.

The nature of the signal mediating these neuron–target interactions is unknown. One possibility is that the signal, or signals, generated by these interactions travels, by retrograde transport, from the axon terminals to its site of action in neuronal perikarya. This signal could be either a molecule produced by target cells and transferred transsynaptically to axon terminals or a molecule produced in neurons which is modified upon entering the axon terminals and returns to the cell body as a signal.

Distinct patterns of gene expression correlate with longitudinal and radial growth of axons

Rapid longitudinal growth of axons correlates with the expression of class II beta tubulin. Radial growth is associated with NF gene expression. The developmental pattern of gene expression (relatively high for class II beta tubulin and low for NF) correlates with the phase of predominantly longitudinal growth of developing axons. Similarly, recapitulation of the developmental pattern during axonal regeneration correlates with the outgrowth of regenerating sprouts. In contrast, the inverse pattern of gene expression (high for NF and low for class II beta tubulin) in maturing neurons correlates with the radial growth of axons. Recapitulation of the developmental pattern after axotomy (i.e., reduced NF expression) correlates with the cessation of radial growth and, instead, with axonal atrophy. Thus, the developmental pattern of gene expression for NF and class II beta tubulin correlates with the longitudinal growth of axons, while the maturational pattern is associated with radial growth.

10.1002/9780470513675.ch12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/9780470513675.ch12 by Sichuan University, Wiley Online Library on [02/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Acknowledgements

Special thanks to Drs John Griffin, Daniel Watson and Thomas Crawford for their helpful suggestions for the manuscript. Aspects of this work were supported by grants from the National Institutes of Health (NS-22849 and NS-20164). The author is the recipient of a Research Career Development Award from the National Institutes of Health (NS-00896).

References

Berthold CH 1978 Morphology of normal peripheral axons. In: Waxman SG (ed) Physiology and pathobiology of axons. Raven Press, New York, p 3–63

Black MM, Smith W 1988 Regional differentiation of the neuronal cytoskeleton with an appendix: diffusion of proteins in the neuron cell body–mathematical approximations and computer simulations. In: Lasek RJ, Black MM (eds) Intrinsic determinants of neuronal form and function. Alan R. Liss, New York, p 463–486

Bond JF, Robinson GS, Farmer SR 1984 Differential expression of two neural cell-specific beta-tubulin mRNAs during rat brain development. Mol Cell Biol 4:1313–1319

Brady ST, Tytell M, Lasek RJ 1984 Axonal tubulin and axonal microtubules: biochemical evidence for cold stability. J Cell Biol 99:1716–1724

Brodal A 1981 Neurological anatomy in relation to clinical medicine, 3rd edn. Oxford University Press, New York

Cragg BG, Thomas PK 1961 Changes in conduction velocity and fibre size proximal to peripheral nerve lesions. J Physiol (Lond) 157:315–327

Czéh G, Kudo N, Kuno M 1977 Membrane properties and conduction velocity in sensory neurones following central and peripheral axotomy. J Physiol (Lond) 270:165–180

Diamond J 1981 The recovery of sensory function in skin after peripheral nerve lesions. In: Gorio A et al (eds) Posttraumatic peripheral nerve regeneration. Raven Press, New York, p 533–548

Dyck PJ, Nukada H, Lais AC, Karnes JL 1984 Permanent axotomy: a model of chronic neuronal degeneration preceded by axonal atrophy, myelin remodeling, and degeneration. In: Dyck PJ et al (eds) Peripheral neuropathy. Saunders, Philadelphia, p 666–690

Farmer SR, Bond JF, Robinson GS, Mbangkollo D, Fenton MJ, Berkowitz EM 1984 Differential expression of the rat beta-tubulin multigene family. In: Borisy GG et al (eds) Molecular biology of the cytoskeleton. Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, p 333–342

Friede RL, Samorajski T 1970 Axon caliber related to neurofilaments and microtubules in sciatic nerve fibers of rats and mice. Anat Rec 167:379–387

Gillespi MJ, Stein RB 1983 The relationship between axon diameter, myelin thickness and conduction velocity during atrophy of mammalian peripheral nerves. Brain Res 259:41–56

Greenman MJ 1913 Studies on the regeneration of the peroneal nerve of the albino rat: number and sectional areas of fibers: area relation of axis to sheath. J Comp Neurol 23:479–513

Hirokawa N 1982 Cross-linker system between neurofilaments, microtubules, and

10.1002/9780470513675.ch12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/9780470513675.ch12 by Sichuan University, Wiley Online Library on [02/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

membranous organelles in frog axons revealed by the quick-freeze, deep-etching method. J Cell Biol 94:129–142

Hoffman PN, Cleveland DW 1988 Neurofilament and tubulin expression recapitulates the developmental program during axonal regeneration: induction of a specific beta tubulin isotype. Proc Natl Acad Sci USA 85:4530–4533

Hoffman PN, Lasek RJ 1975 The slow component of axonal transport: identification of the major structural polypeptides of the axon and their generality among mammalian neurons. J Cell Biol 66:351–366

Hoffman PN, Lasek RJ 1980 Axonal transport of the cytoskeleton in regenerating motor fibers: constancy and change. Brain Res 202:317–333

Hoffman PN, Griffin JW, Price DL 1984 Control of axonal caliber by neurofilament transport. J Cell Biol 99:705–714

Hoffman PN, Griffin JW, Gold BG, Price DL 1985a Slowing of neurofilament transport and the radial growth of developing nerve fibers. J Neurosci 5:2920–2929

Hoffman PN, Thompson GW, Griffin JW, Price DL 1985b Changes in neurofilament transport coincide temporally with alterations in the caliber of axons in regenerating motor fibers. J Cell Biol 101:1332–1340

Hoffman PN, Cleveland DW, Griffin JW, Landes PW, Cowan NJ, Price DL 1987 Neurofilament gene expression: a major determinant of axonal caliber. Proc Natl Acad Sci USA 84:3472–3476

Hursh JB 1939 Conduction velocity and diameter of nerve fibers. Am J Physiol 127:131–139

Julien J-P, Meyer D, Flavell D, Hurst J, Grosveld F 1986 Cloning and developmental expression of the murine neurofilament gene family. Mol Brain Res 1:243–250

Kuno M, Miyata Y, Munoz-Martinez EJ 1974 Properties of fast and slow alpha motoneurones following motor reinnervation. J Physiol (Lond) 242:273–288

Lasek RJ 1981 The dynamic ordering of neuronal cytoskeletons. Neurosci Res Program Bull 19:7–32

Lasek RJ, Hoffman PN 1976 The neuronal cytoskeleton, axonal transport and axonal growth. Cold Spring Harbor Conf Cell Proliferation 3:1021–1049

Lewis SA, Cowan NJ 1985 Genetics, evolution and expression of the 68,00-mol-wt neurofilament protein: isolation of a cloned cDNA probe. J Cell Biol 100:843–850

Martinez AJ, Friede RL 1970 Changes in nerve cell bodies during the myelination of their axons. J Comp Neurol 138:329–338

Morris JR, Lasek RJ 1982 Stable polymers of the axonal cytoskeleton: the axoplasmic ghost. J Cell Biol 92:192–198

Peters A, Vaughn JE 1967 Microtubules and filaments in the axons and astrocytes of early postnatal optic nerves. J Cell Biol 32:113–119

Ramón y Cajal S 1928 Degeneration and regeneration of the nervous system. Hafner, New York

Sullivan KF, Cleveland DW 1986 Identification of conserved isotype-defining variable region sequences for four vertebrate beta tubulin polypeptide classes. Proc Natl Acad Sci USA 83:4327–4331

Weiss P, Edds MV Jr, Cavanaugh M 1945 The effect of terminal connections on the caliber of nerve fibers. Anat Rec 92:215–233

Weiss PA, Mayr R 1971 Organelles of neuroplasmic ('axonal') flow: neurofilaments. Proc Natl Acad Sci USA 68:846–850

Wujek JR, Lasek RJ 1983 Correlation of axonal regeneration and slow component *b* in two branches of a single axon. J Neurosci 3:243–251

Yamada KM, Spooner BS, Wessels NK 1971 Ultrastructure and function of growth cones and axons of cultured nerve cells. J Cell Biol 49:614–635.

10.1002/9780470513675.ch12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/9780470513675.ch12 by Sichuan University, Wiley Online Library on [02/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

DISCUSSION

Kernell: You found a strong correlation between axon diameter and the number of neurofilaments and you observed no very good correlation between axon diameter and the number of microtubules. Does that mean that the amount of microtubules per cross-sectional area might vary with the functional state of the neuron? This would imply that the capacity for fast axonal transport might be changeable.

Hoffman: I agree. It is quite possible that different populations of axons with comparable cross-sectional areas may differ in the density of microtubules.

Kernell: Do you see any systematic difference between smaller and larger alpha axons, or between alpha and gamma axons, in the density of microtubules per cross-sectional area?

Hoffman: We haven't looked specifically at larger and smaller alpha axons. If we compare large and small motor fibres in the ventral root, which are presumably alphas and gammas, the density is higher for the smaller fibres (Friede & Samorajski 1970). Using our relatively crude scheme I am not sure how we would differentiate large and small alphas. Would you just use cross-sectional area as the criterion? We could do that.

Kernell: It would be interesting to do.

Buller: However, there are variations in diameter along the length of alpha fibres.

Kernell: Yes, but for these comparisons between large and smaller alpha axons it would primarily be of interest to rank the nerve fibres according to size (e.g., mean diameter or cross-sectional area). Absolute sizes would presumably be less important.

Lowrie: You described the atrophy of the axon after axotomy followed by recovery and regeneration. We have been looking at the effect of crushing the sciatic nerve in young rats in the first week after birth (Lowrie et al 1982, 1987). After crushing at five days of age we see regeneration of apparently all the motoneurons back to the muscle, and we get some recovery of the muscle, but we have been struck by the fact that the sciatic nerve, both proximal and distal to the point of crush, remains rather thin. We haven't measured the axons, but my guess is that axonal diameters are reduced. Have you any comment on that, in relation to your findings on interaction with the target?

Hoffman: The induction of neurofilament expression in rat sensory neurons between five and ten days of age correlates with the onset of radial growth in myelinated axons (N.A. Muma & P.N. Hoffman, unpublished work). It is conceivable that axonal injury prevents this induction, thereby impairing radial growth.

Lowrie: We find differences both in the muscle and in the motoneurons. There is a change in the size distribution of the motoneurons, and, in the fast

10.1002/9780470513675.ch12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/9780470513675.ch12 by Sichuan University, Wiley Online Library on [02/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

muscles at least, a reduction in the size of the motor units. Both could be related to reduction in the size of the axon.

Nadal-Ginard: Are the changes in neurofilament mRNA and tubulin mRNA transcriptionally regulated, or is it a half-life regulator?

Hoffman: In this *in vivo* system we are limited in the number of cells that we obtain, so we cannot look at transcriptional levels. We can only guess as to whether we are looking at increased turnover of neurofilament message or reduced transcription.

Nadal-Ginard: You emphasized the kinetics of the changes of different steady-state levels for these two mRNAs. Do you know whether the neurofilament mRNA has a longer half-life than tubulin mRNA? Would a difference in half-life explain the asynchronous behaviour of the two?

Hoffman: We have no information on the half-life or the transcriptional rates of either mRNA in this system.

Nadal-Ginard: With an *in vivo* system, however, you could look at half-life by inhibition of RNA synthesis.

Zak: There is a potential hazard of the assay for measuring the quantity of messenger RNA. If the data are expressed per unit of RNA and you find for example that the abundance of mRNA is half, and the regenerating tissue has twice as great a concentration of RNA, the total amount of mRNA will not change. I would interpret some of your *in situ* hybridizations as showing less change than the blotting assay.

Hoffman: It is our impression that the RNA blots are more sensitive than *in situ* hybridization for detecting changes in mRNA levels. A 10-fold reduction in neurofilament mRNA levels measured on RNA blots corresponds to a threefold reduction in grain density over neurons examined by *in situ* hybridization (Hoffman & Cleveland 1988).

Zak: From the sequence of your probe, wouldn't it be advisable to design a generic probe which would tell you the total amount of mRNA and use it as a reference for the expression of specific mRNA?

Hoffman: Yes, when we examined the expression of beta amyloid in regenerating sensory neurons we found essentially no change in mRNA levels (E.H. Koo, N.A. Muma, P.N. Hoffman & D.L. Price, unpublished work). This suggests that there is little, if any, change in the total RNA content of these neurons.

Oppenheim: Have experiments been done to see whether the target-derived factor is NGF? This would indicate whether you are dealing with a sensory neuron.

Hoffman: We don't know, and we should investigate this, using NGF or an antibody to NGF, or both.

Van Essen: Much of your measured response may be the active response to the obligation to regenerate that these nerves have, but part of it may be the actual disconnection. Has anyone looked at the effects of blocking neuro-

10.1002/9780470513675.ch12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/9780470513675.ch12 by Sichuan University, Wiley Online Library on [02/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

muscular transmission, to see what changes in either axonal diameter or cytoskeletal composition occur?

Hoffman: Blocking neuromuscular transmission with botulinum toxin produces somatofugal axonal atrophy in motor neurons (B.G. Gold, J.W. Griffin, A. Pestronk, P.N. Hoffman, E.F. Stanley & D.L. Price, unpublished work). We are currently examining cytoskeletal gene expression in this system (T.O. Crawford, J.W. Griffin & P.N. Hoffman, unpublished work).

Henderson: I am interested in what is really happening to the neurofilaments when, for instance, a nerve terminal dies back or is removed during the regression of polyneuronal innervation. Initially, there is a stack of neurofilaments which give a certain diameter to the axon. Are they maintained as a stack while being degraded by the calcium-dependent protease from the end?

Hoffman: This is not known.

Kernell: How does this system work at branchings? Do you preserve the same total number of neurofilaments throughout the whole branching tree of an axon?

Hoffman: We have not examined branch points, but it is clear that neurofilament number is not conserved along unbranched portions of myelinated nerve fibres. In large-calibre fibres the axonal cross-sectional area may be as much as 10-fold greater at internodes than at nodes. Since neurofilament density is comparable at nodes and internodes, neurofilament number is substantially greater along internodal segments (Berthold 1978). This suggests that there must be a reorganization of the cytoskeleton as neurofilaments are transported through nodes.

There also appears to be reorganization of axonal neurofilaments during radial growth of myelinated nerve fibres. Examination of transport kinetics reveals that there is a continuous decline in the velocity of pulse-labelled neurofilament protein with increasing distance along the nerve (Hoffman et al 1985a). Therefore, neurofilaments enter each region of an axon (e.g., an internodal segment) slightly faster than they leave, leading to a net accumulation of neurofilaments over time. We believe that this contributes directly to the radial growth of myelinated axons, a process which occurs continuously during the first six months of postnatal development in peripheral nerve fibres of the rat (Hoffman et al 1985a).

Kernell: Can one see neurofilament 'bricks' lying around in the axon?

Hoffman: Our information is quite limited here. We do not know whether neurofilaments are relatively short, or whether they are infinitely long and folded back on themselves within each internodal segment (see Hoffman et al 1984).

Van Essen: When you post-label the neurofilaments, how sharp is the region of labelling as it migrates down?

Hoffman: The peak is initially sharp and spreads as it moves along the nerve (Hoffman et al 1983).

10.1002/9780470513675.ch12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/9780470513675.ch12 by Sichuan University, Wiley Online Library on [02/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Sanes: Is there any manipulation that keeps the production of neurofilaments up? If so, how is axon calibre regulated? Is there a limit?

Hoffman: Although the available evidence indicates that the level of neurofilament gene expression is a major determinant of axonal calibre in large myelinated nerve fibres, an unresolved issue is the role of target cells in regulating neurofilament gene expression. One possibility is that the level of neurofilament expression is specified by trophic interactions between neurons and their targets. Alternatively, both the level of neurofilament expression and target preference could be intrinsic properties of neurons. In the latter case, the target would play a permissive (rather than deterministic) role in regulating neurofilament gene expression; that is, target cell interactions would be required for expression to occur, but the level of expression would be determined by other factors.

Walsh: Can you detect temporal changes in the expression of the three neurofilament protein subunits in your lesion model?

Hoffman: No. In axonal transport experiments we found that levels of radioactivity in each of the three neurofilament proteins are reduced proportionately during regeneration (Hoffman et al 1985b). More recently, using cDNA probes, we found comparable reductions in mRNA levels for all three neurofilament proteins. Thus, it appears that the expression of these proteins is coordinated.

Kernell: Are dendrites regulated in the same way as axons?

Hoffman: We haven't looked. In sensory neurons there is however a correlation between neurofilament expression and cell body size, as well as axonal diameter. In the axotomy model, when the axons shrink, the cell bodies are also smaller. That was shown by Cavanaugh (1951) and we found the same. Of course, there are more things in the cell body than neurofilaments, so I don't want to say that neurofilaments control cell body size, but they contribute.

Kernell: There is also a larger total sum of dendritic stem diameters going out from a large than from a smaller motoneuronal cell body (e.g. Zwaagstra & Kernell 1981).

Hoffman: Yes, and there is evidence that the large dendrites also have neurofilaments, and that the neurofilament density is comparable in large dendrites and in large axons, at about $100/\mu m^2$.

Zak: During branching, the diameter of the axon gets smaller. Is there any increased vulnerability of the axons of small diameter?

Hoffman: We have no data on this.

References

Berthold CH 1978 Morphology of normal peripheral axons. In: Waxman SG (ed) Physiology and pathobiology of axons. Raven Press, New York, p 3-63

Cavanaugh MW 1951 Quantitative effects of the peripheral innervational area on nerves and spinal ganglion cells. J Comp Neurol 94:181–219

10.1002/9780470513675.ch12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/9780470513675.ch12 by Sichuan University, Wiley Online Library on [02/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Friede RL, Samorajski T 1970 Axon caliber related to neurofilaments and microtubules in sciatic nerve fibers of rats and mice. Anat Rec 167:379–387

Hoffman PN, Cleveland DW 1988 Neurofilament and tubulin expression recapitulates the developmental program during axonal regeneration: induction of a specific beta tubulin isotype. Proc Natl Acad Sci USA 85:4530–4533

Hoffman PN, Lasek RJ, Griffin JW, Price DL 1983 Slowing of the axonal transport of neurofilament proteins during development. J Neurosci 3:1694–1700

Hoffman PN, Griffin JW, Price DL 1984 Neurofilament transport in axonal regeneration: implications for the control of axonal caliber. In: Elam JS, Cancalon P (eds) Axonal transport in neuronal growth and regeneration. Plenum Press, New York, p 243–260

Hoffman PN, Griffin JW, Gold BG, Price DL 1985a Slowing of neurofilament transport and the radial growth of developing nerve fibers. J Neurosci 5:2920–2929

Hoffman PN, Lasek RJ, Griffin JW, Price DL 1985b Changes in neurofilament transport coincide temporally with alterations in the caliber of axons in regenerating motor fibers. J Cell Biol 101:1332–1340

Hoffman PN, Koo EH, Muma NA, Griffin JW, Price DL 1988 Role of neurofilaments in the control of axonal caliber in myelinated nerve fibers. In: Lasek RJ, Black MM (eds) Intrinsic determinants of neuronal form. Alan R Liss, New York, p 389–402

Lowrie MB, Krishnan S, Vrbová G 1982 Recovery of slow and fast muscles following nerve injury during early post-natal development in the rat. J Physiol (Lond) 331:51–66

Lowrie MB, Krishnan S, Vrbová G 1987 Permanent changes in muscle and motoneurons induced by nerve injury during a critical period of development of the rat. Dev Brain Res 31:91–101

Zwaagstra B, Kernell D 1981 Sizes of soma and stem dendrites in intracellularly labelled alpha-motoneurons of the cat. Brain Res 204:295–309

10.1002/9780470513675.ch12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/9780470513675.ch12 by Sichuan University, Wiley Online Library on [02/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

General discussion II

Plasticity, specialization and adaptability

Buller: It still remains unknown whether neurons are predetermined and will specifically innervate only particular types of muscle, or whether motor axons, having reached the target, are informed in some way by the target cell how their cell bodies ought to develop. I don't know the opinion of those currently in the field, but it seems to me that since the Konstanz symposium (Pette 1980) there has been a move away from 'plasticity' (the idea that motoneurons are not predetermined) back to the earlier views stressing the specificity of motoneurons. Would anyone like to comment on that? At the beginning of the symposium, I felt that Dirk Pette was constraining muscle fibres to work only within fixed limits.

Pette: In fact, the more we study populations of muscle fibres, the more we detect diversity *and* specialization. For example, certain troponinT subunits display muscle-specific distribution patterns in hindlimb, masticatory or tongue muscles (Moore et al 1987). In addition, within the same muscle, fibres may be histochemically classified as identical, yet microbiochemical analyses indicate discrete differences in the expression of certain proteins (Pette & Spamer 1986, Staron & Pette 1987a,b). These variations may reflect specialization brought about by the functional demand, but they also reflect the dynamic state of muscle fibres—that is to say, the plasticity of muscle.

Vrbová: We should perhaps try to define plasticity; as a term it is rather diffuse. What I understand by plasticity is the ability of a structure, whether a neuron or nervous system or muscle, to adapt to functional demands. And because we are dealing with systems made up of several components, we can talk about plasticity only in relation to particular components and a particular function. This is why the myosin molecule and the expression of its various isoforms will probably be related to its function—that is, to the mechanical conditions under which it works and to the amount of work that it does. It is important to emphasize that plasticity is not just an overall response of the cell. It can be coordinated, but it need not be.

Buller: That is what I was suggesting, that we are now constraining plasticity, a concept that went through a period where it was believed that everything can do almost anything.

Kernell: We need a term that implies neither limitless plasticity, nor that everything is fixed: something like 'controlled adaptability'.

Rubinstein: Some of the difference in emphasis between the Konstanz symposium and this one is that then we talked about what we could do to cells, to

10.1002/9780470513675.ch12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/9780470513675.ch12 by Sichuan University, Wiley Online Library on [02/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

make them different, with papers on regeneration, cross-innervation, and the effects of chronic stimulation. We are now asking what normally occurs during development and during disease processes. We have discovered that this isn't quite what we thought. At the time of the earlier symposium, we saw muscle cells as blank slates and we thought that exogenous factor(s) would come in and determine what each fibre became. We know now that this is not true.

Zak: It may be useful to think about these problems in terms of differentiation programmes. Perhaps we should start from the fully undifferentiated cell (the premyoblast). These cells go through several stages of differentiation, but in each case there is a programme which is 'possible' and another which is prohibited. It is a question of having certain 'windows' for phenotypic decisions, and within each window there is a certain adaptability that cannot be exceeded.

Gordon: I like that way of looking at it, and my paper will be concerned with what one could consider to be pathology but I think is plasticity, after reinnervation. To some extent, regeneration involves dedifferentiation or reversion, and is similar to development. The window concept may be important here. Much of the response to injury is a limited and also an adaptive dedifferentiation. It may not go back to the earliest stages of development. Many of the specific cues are lost in the plasticity of the adult. We have evidence (Gordon et al, this volume) that in regeneration the process of 'getting back' is a non-specific one, but the residual plasticity of the neuromuscular system allows for recovery of the original properties.

Buller: In other words, some doors or windows that are gone through during development are closed, and you cannot go back by the same route.

Gordon: Exactly!

Van Essen: The other point to remember is that these windows seem narrowed or constrained only in relation to the infinite initial possibilities. The number of decisions that can still be taken within a window is much larger than we might have realized earlier, and to choose among 10^9 or so myosin isoform possibilities, or to choose pathways with as much specificity as we now know exists, in some respects signifies that a vast amount of plasticity is present.

Nadal-Ginard: As we learn more about the neuromuscular system, it is logical that it seems more complex and more specialized. One issue that is confusing is the fact that we are detecting more specialization, but that doesn't necessarily mean that we are constraining the plasticity. Dr Van Essen just said that the neuromuscular system may be more plastic than we expected, because there are many more decisions to be made than we thought. The question is what these ideas about 'windows' and 'programmes' mean in biochemical terms. For neuromuscular gene expression, we need to know which phenotypes are allowed and which are not allowed, and why.

One way of addressing these issues is to look for interspecific differences in gene expression. Closely related animal species may have different develop-

10.1002/9780470513675.ch12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/9780470513675.ch12 by Sichuan University, Wiley Online Library on [02/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

mental programmes, especially for the contractile proteins. What does this mean biochemically? Does it mean that these developmental differences are encoded in the sequence of the structural genes, or depend on other regulatory genes? In other words, are these differences *cis* or *trans* regulated? For some of the myosin genes we know that certain differences among mammalian species are *trans* regulated.

Buller: Are there examples of genes that can be turned on in development but cannot be turned on again in the adult?

Nadal-Ginard: There are examples of developmentally dependent genes, but the contractile protein genes that have been studied so far are not of this type, but are reversibly induced. The number of examples of genes that cannot be turned on again is more limited than the literature would suggest. In some instances the cell type that was expressing the gene early in development has disappeared (e.g. by programmed cell death), and this is why the gene appears not to be re-expressed.

Pette: Chronic stimulation experiments disclose pronounced species-specific differences in responsiveness. We have compared the effect of 10 hours per day of indirect (10Hz) stimulation of the tibialis anterior (TA) muscle in mouse, guinea-pig, rat and rabbit (Simoneau & Pette 1988). To our surprise, the same stimulus pattern evoked very different responses in these small mammals. Stimulation brought about only slight increases in enzyme activities of aerobic-oxidative metabolism in mouse TA, but produced several-fold increases in rabbit TA. Rat and guinea-pig were intermediate (Simoneau & Pette 1988). The question arises whether the mouse, being a small and active animal, already has maximally expressed these enzymes, whereas the sedentary rabbit normally operates at a submaximal level of expression.

Vrbová: It is difficult to know whether a cell has lost the ability to express a new gene,or whether we do not know how to switch it on. For example, if you look at the cable properties of membranes of slow-tonic and fast-twitch muscle fibres, once they are established you can't alter them and they remain recognizably different. This doesn't mean that if one can find the right stimulus, the potential is not there.

Gordon: One of the best examples of that is the growth-associated proteins (GAPs). Growing neurons in the central nervous system all express these proteins. Central neurons don't regenerate; this is because they lack the right environment in which to grow. If they are induced to grow into the Schwann cell tubes of the peripheral nervous system they can re-express GAPs. So it is the induction of the protein, or the inductive stimulus, which may be lost, or appear to be lost.

Vrbová: It could be our limitations that prevent us from finding the appropriate stimulus.

Zak: An example of a 'one-way' gene is the embryonic haemoglobin, which is never re-expressed in the adult.

10.1002/9780470513675.ch12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/9780470513675.ch12 by Sichuan University, Wiley Online Library on [02/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Nadal-Ginard: This is because the cells that make it, in the liver, have disappeared in the adult. As an example of the complexity of the regulatory system, Thy-1 antigen expression in humans and mice is very different. This difference is encoded in the sequence of the human and mouse genes. When the human gene is introduced into mice in transgenic animals it retains the human pattern of expression. This is not the case for myosin. The human has a cardiac myosin pattern of expression that is very different from rat or mouse. This different programme is not encoded in the myosin genes. Human genes in mouse cells behave like mouse genes and vice versa. So here are two examples of species-specific phenotypic differences that are regulated differently. Thy-1 seems to be *cis* regulated. The Thy-1 gene dictates the pattern of expression independently of the species, whereas myosin is *trans* regulated. The gene does not have the information for the species-specific difference, but the cell does. A human cell imposes a pattern of expression on a human as well as a mouse myosin gene introduced into it, and vice versa.

Eisenberg: The question is whether muscle is able to go through all the gene programmes again, or whether we are unable to trigger the correct response. When we study an adult cell (as we do with our invasive techniques) we are not producing all the events that happened originally in development. A physiologist looks at properties that we cannot recapitulate—the kind of work that a muscle fibre does, the load against which it is now working. In the adult, the connective tissue is in place already and the tendon ends are connected up. In some pathological situations, for example where overloading of chicken ALD is extreme, you can get a population of migrating satellite cells that form new myotubes which express a myosin characteristically found only during embryonic development (Kennedy et al 1988).

Zak: On the potential effect of the microenvironment, it must be an extremely complicated situation. Each gene family of muscle proteins has a completely different expression programme and a different regulatory programme (Swynghedauw 1986). One kind of stimulus affects one gene family and may cause opposite changes in another gene family.

Oppenheim: With regard to the issue of species differences, we usually study highly domesticated species (as in Dr Pette's studies of rat, rabbit, and other species) where selection pressures may have loosened. Consequently, the differences that we are seeing may not have much biological significance. The kinds of species differences that would be more biologically interesting are those expressed in wild-type or non-domesticated animals. Until this study is done it seems premature to argue that species differences in muscle properties of highly domesticated animals are biologically meaningful.

References

Gordon T, Bambrick L, Orozco R 1988 Comparison of injury and development in the neuromuscular system. In: Plasticity of the neuromuscular system. Wiley, Chichester (Ciba Found Symp 138) p 210–226

10.1002/9780470513675.ch12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/9780470513675.ch12 by Sichuan University, Wiley Online Library on [02/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Kennedy JM, Eisenberg BR, Reid SK, Sweeney LJ, Zak R 1988 Nascent muscle fiber appearance in overloaded chicken slow-tonic muscle. Am J Anat 181:203–215

Moore GE, Briggs MM, Schachat FH 1987 Patterns of troponin T expression in mammalian fast, slow and promiscuous muscle fibres. J Muscle Res Cell Motil 8:13–22

Pette D (ed) 1980 Plasticity of muscle. Walter de Gruyter, Berlin/New York

Pette D, Spamer C 1986 Metabolic properties of muscle fibers. Fed Proc 45:2910–2914

Simoneau JA, Pette D 1988 Species-specific effects of chronic nerve stimulation upon tibialis anterior muscle in mouse, rat, guinea pig and rabbit. Pfluegers Arch Eur J Physiol 412:86–93

Staron RS, Pette D 1987a The multiplicity of myosin light and heavy chain combinations in histochemically typed single fibres of rabbit soleus muscle. Biochem J 243:687–693

Staron RS, Pette D 1987b The multiplicity of myosin light and heavy chain combinations in histochemically typed single fibres of rabbit tibialis anterior muscle. Biochem J 243:695–699

Swynghedauw B 1986 Development and functional adaptation of contractile proteins in cardiac and skeletal muscles. Physiol Rev 66:710–770

10.1002/9780470513675.ch12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/9780470513675.ch12 by Sichuan University, Wiley Online Library on [02/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License